D 132 | Stress the units, not the forecast

PRIVATE FICTIONAL tabletop. No real budget, contract, vendor onboarding, acquisition campaign, participant testing, account change or spend.

Acquisition chain	100000 impressions at CPM 200: spend 20000, clicks 2000 at 2% CTR,100 distinct new customers at 5% click conversion, CAC 200. CPM 320 alone yields CAC 320; CTR 1.6% and conversion 4% gives 64 customers and CAC 500. Model expectations can be fractional after edits, not actual people.
Customer weights	70% of acquired customers at CAC 200 and 30% at 300 gives 230. Keep those CUSTOMER shares and change Meta CAC 320 ->314. Spend shares are not customer shares; changing channel mix requires new weights.
Monthly cash bridge	Lakh INR:10 opening+5 collected sales less paid variable product/fulfilment costs-2.3 marketing-3 fixed-1 additional working-capital build-.5 principal=8.2. The 5 excludes these later deductions; no fulfilment deducted twice. Not profit or intramonth cash minimum.
Separate option horizon	Thousands INR, one horizon, no discount/tax: unprotected 200 p, standby option 20+100 p, full redundancy 80. Minimum-cost path: unprotected below.2; option between.2/.6; redundancy above.6; ties at boundaries. Unprotected/redundancy tie.4 is not global optimum.
Capacity and loop	Option assumes exercise 60 plus residual 40 on failure and capacity sufficient. Added required 100/available 100 units passes only quantity; lead time, shared failure, service quality and legal rights still open. Failures -> refunds/backlog -> cash pressure -> capacity cuts -> failures is a positive reinforcing hypothesis, with delays.
Evidence and authority	Scenarios are narratives, not exhaustive probability bins. Toyp is an assumed probability in a separate expected-cost model, not inferred from scenarios. Paper triggers/owners do not approve spend. Sources written only, no full AV/interactive/native parity/independent learner/account/legal or overall 10 claim.

Inputs | Each model keeps its own units

Yellow controls fictional. Probability applies only to separate single-horizon option toy, not scenario narratives. A positive quantity flag never supplies timing, contracts or permission.

Impressions	100000.0000
Base CPM, INR/1000 impressions	200.0000
Shock CPM, INR/1000 impressions	320.0000
Base click-through fraction	0.0200
Base purchase per click fraction	0.0500
Shock click-through fraction	0.0160
Shock purchase per click fraction	0.0400

Meta share of acquired customers, fraction	0.7000
Base Meta CAC, INR/customer	200.0000
Other CAC, INR/customer	300.0000
Shock Meta CAC, INR/customer	320.0000

Opening cash, lakh INR	10.0000
Collected net of paid variable costs, lakh	5.0000
Marketing cash, lakh	2.3000
Fixed cash costs, lakh	3.0000
Additional working-capital build, lakh	1.0000
Principal paid, lakh	0.5000

Failure probability, one horizon	0.4000
Unprotected failure loss, thousand INR	200.0000
Option standby, thousand INR	20.0000
Option exercise on failure, thousand INR	60.0000

Inputs | Each model keeps its own units

Yellow controls fictional. Probability applies only to separate single-horizon option toy, not scenario narratives. A positive quantity flag never supplies timing, contracts or permission.

Option residual loss on failure, thousand INR	40.0000
Redundancy cost, thousand INR	80.0000

Required alternative capacity, units	100.0000
Available alternative capacity, units	100.0000

Acquisition inputs valid?	1.000000
Customer blend inputs valid?	1.000000
Cash inputs valid?	1.000000
Option inputs valid?	1.000000
Capacity inputs valid?	1.000000

Cash collected less variable costs may be negative. Zero CTR/conversion permits spend/click outputs, but CAC is undefined with no modeled customers. Input gates isolate models; static prose remains baseline and must be revised after edits.

Acquisition | CPM is not CAC

All modeled purchases are distinct new customers in this toy. Spend INR, counts per modeled campaign, CAC INR/customer. Conversion is conditional on clicks, not all impressions.

Base spend, INR	20000.000000
Base clicks	2000.000000
Base new customers	100.000000
Base CAC, INR/customer	200.000000
Shock spend, INR	32000.000000
CPM-only customers, unchanged rates	100.000000
CPM-only CAC	320.000000
Combined-shock clicks	1600.000000
Combined-shock new customers	64.000000
Combined-shock CAC	500.000000

Base 20000/100=200. CPM-only 32000/100=320. Combined 32000/64=500. New-customer assumption is not attribution evidence or incrementality. Repeat purchasers/duplicates/refunds/organic overlap need actual cohort reconciliation. No campaign executed.

Blend | Customer shares are the weights

A separate fixed-customer-mix calculation, not automatically equal to the acquisition model. No channel population or spend weights supplied.

Base blended CAC, INR/customer	230.000000
Shock blended CAC, INR/customer	314.000000

Baseline.7*200+.3*300=230; shock.7*320+.3*300=314. Applying 60% to the whole 230 gives 368 and is wrong under fixed customer weights. If weights instead represent spend shares, a different denominator calculation is needed. No mix stability inferred.

Cash | One deduction per cash item

Lakh INR/month. Collected sales less paid variable product/fulfilment costs is already 5. Working-capital build is additional and not already deducted. No other flows/tax assumed for this toy.

Net monthly cash change, lakh	-1.800000
Closing endpoint cash, lakh	8.200000
Endpoint decline, lakh	1.800000

10+5-2.3-3-1-.5=8.2; net change-1.8. Principal is cash outflow, not operating expense; working capital is not deducted twice. Fixed-cost absorption is an accounting allocation, not another cash payment. No profit, runway extrapolation, debt covenant or intramonth minimum inferred.

Options | Compare all three, including ties

Thousands INR, single horizon, no discount/tax. Failure probability is assumed, not frequency estimated from scenario labels. Redundancy completely avoids stated failure loss only by assumption.

Unprotected expected cost	80.000000
Option expected cost	60.000000
Redundancy cost	80.000000
Lowest expected cost, not unique choice	60.000000

Unprotected-option algebraic crossing p	0.200000
Option-redundancy algebraic crossing p	0.600000
Unprotected-redundancy crossing p	0.400000

Atp.4 costs 80/60/80: option minimum 60 despite other pair tying 80. Baseline minimum path unprotected p<.2, option.2<p<.6, redundancy p>.6; equal minima at.2/.6. Edited crossings may fall outside 0..1 or reverse rankings: inspect all costs, do not reuse baseline regions. Parallel/coincident lines have no unique crossing. Expected minimum is not worst-case survival, risk appetite, affordability or permission.

Capacity | Quantity alone is not resilience

Added independent capacity toy, not source evidence. Required/available alternative units must refer to same event window, quality and location. Does not automatically change original residual-loss assumption.

Available minus required capacity, units	0.000000
Quantity requirement met?1=yes	1.000000
Uncovered requirement, units	0.000000

100 available-100 required=0,quantity passes at equality. Lower availability 80 leaves 20 uncovered; full avoidance/residual 40 assumptions then need revision, not blind expected-cost comparison. Check lead time, readiness, shared upstream failures, contracted availability and approvals separately. No supplier tested or onboarded.

Loop | Close the causal hypothesis

Signed links are hypotheses, not regression coefficients or proven dynamics. Each link compares a change holding other influences fixed. Delays can change timing and observed behavior.

1 | Service failures -> backlog	Positive: more failed services can add unresolved refunds/backlog. Delay: detection and claim processing. Some failures do not produce refunds; scope and observation window must be defined.
2 | Backlog -> cash pressure	Positive: more refund/backlog obligations can increase cash pressure. Delay: settlement and refund payment. Cash pressure means shortage pressure, not cash balance, so sign is not reversed.
3 | Cash pressure -> capacity cuts	Positive: higher cash pressure may prompt cuts to staffing/fulfilment capacity. Delay: management decisions, contract terms and staffing changes. Could fail if buffers or financing prevent cuts.
4 | Capacity cuts -> failures	Positive: less service capacity can increase future failures, closing the loop to step 1. Delay: workload accumulation. Four positive links imply reinforcing polarity, not guaranteed exponential collapse.
Alternative variable names	If using capacity LEVEL instead of CUTS, cash pressure ->capacity is negative and capacity->failures negative: two negative links still yield reinforcing polarity. Do not label an open chain a loop or infer system behavior from polarity alone.
Intervention is not permission	A proposed buffer, queue control or reversible capacity action needs evidence, timing, resource/contract/customer authority and trade-offs. LOOPY landing text only; no executed animation, quantitative causal estimate or calibrated simulation.

Scenarios | Narratives and decision gates

No exhaustive scenario set or empirically estimated probabilities. Tabletop only; fictional roles are not people who accepted responsibility.

Do not collapse drivers	Acquisition CPM/CTR/conversion, fulfilment failure, collection delay and alternative capacity may interact. Test plausible combinations and contrary evidence, not only one optimistic/pessimistic slider or four boxes that claim all futures.
Base and revision	McKinsey 2009 recommends a base/central case, contingencies and dynamic revision; its at-least-four rule is a practitioner heuristic. OECD 2025 foresight does not predict a single most-likely future. Different purpose, not permission to manufacture probabilities.
Paper trigger/window	Example only: defined seven-day service-failure rate above 5% on at least 100 eligible orders prompts private review within one day. Not a current alert, real threshold policy, appointment or authority grant. Record denominator, window, data source and false-alarm risk.
Fictional owner/fallback	Operations role diagnoses backlog/capacity; finance role reconciles cash/commitments; sponsor role decides any allowed scope. If data/capacity/authority missing, hold affected action and retain a paper alternative. No accepted real owners.
Before execution	Check permitted vendor terms, readiness/lead time/shared risks, full cash/commitments, privacy/customer effects and specific budget/account authority. Trigger satisfaction and expected value do not supply any of these.
Learn from tabletop	Record observed misunderstanding, correction, strongest objection and delayed transfer only if performed. Authored formula tests do not establish learner behavior or business continuity readiness. Actual records remain blank.

Evidence | Written sources, bounded claims

Refreshed 13 November 2026. Consultancies, government/intergovernmental guidance, teaching glossary and tool-maker landing have different purposes; not empirical validation of this business.

McKinsey2009	https://www.mckinsey.com/capabilities/strategy-and-corporate-finance/our-insights/the-use-and-abuse-of-scenarios - full returned written article by Charles Roxburgh read. Breadth, tails, revision, base/contingencies and practitioner heuristics; historical forecasts/examples not current fact or universal laws.
Ready.gov2026	https://www.ready.gov/business-continuity-plan - written landing updated25March2026, team/plan/test sequence and video descriptions. Planning Suite no longer supported/available. Supporting manuals/videos not inspected or executed; not Indian compliance or real team readiness.
OECD2025	https://www.oecd.org/en/publications/managing-emerging-critical-risks_1f9858ea-en.html - returned abstract/executive summary only, four-country public-risk governance, not full report or retailer proof. Toolkit PDF selected foresight/nonprediction passage only; not full methods/case studies/figures.
Toolkit URL	https://www.oecd.org/content/dam/oecd/en/publications/reports/2025/01/foresight-toolkit-for-resilient-public-policy_9ad1cd60/bcdd9304-en.pdf - OECD2025 Strategic Foresight Toolkit for Resilient Public Policy. Long-term public-policy foresight differs from monthly firm stress assumptions; no OECD endorsement or source asset copied.
MIT glossary	https://ocw.mit.edu/courses/15-988-system-dynamics-self-study-fall-1998-spring-1999/c8a435c4649cb265aadf0b8f718c9f9f_glossary.pdf - selected written causal-loop/polarity/delay/unit definitions only, not full course/PDF diagrams or calibrated dynamics.
LOOPY and limits	https://ncase.me/loopy/ - full returned written landing only, not interactive simulation executed. Six source URLs are not six independent efficacy replications. Native QA is authored LibreOffice, not Excel/Sheets parity, independent learner/account/legal or overall10.

Actual work | No simulated continuity readiness

Record only work actually performed under authority. No private customer records, vendor contact or real campaign data required.

Date/version	Units/denominator	Cash correction	Loop objection	Option/capacity	Reviewer finding	Delayed transfer

Actual work remains blank. Record only what occurred: dates, reasoning, error/source/correction, strongest objection and a new delayed transfer. No invented learner results, sponsor acceptance, actual budget release or signed rights.

Failure log | Authored execution only

Observed native tests are added only after execution. Not actual learner mistakes, independent human audit, Excel/Sheets parity, real authority or overall 10/10.

Controlled test	Observed authored outcome
Independent baseline arithmetic	PASS:30 baseline formulas independently recomputed.
Changed acquisition/customer mix	PASS:5 acquisition/3 fixed-customer-mix changed sets.
Changed cash/option probability	PASS:4 cash/4 probability sets including boundary ties.
Changed option costs/capacity	PASS:5 option-cost/3 capacity changed sets.
Invalid controls and isolation	PASS:16 invalid controls and unaffected model isolation.
Broken-reference explicit holds	PASS:6 broken references produce explicit calculation errors.
Restored baseline/clean caches	PASS:24 changed sets+restored baseline; clean saved caches.
Actual/reviewer/transfer blank	PASS:actual learner/reviewer/delayed transfer remain blank.

Only the original synthetic course exercise at https://top1os.dev/ supplies the case figures. No new empirical reading, real participants, company records, Indian legal clearance, actual commitments or source completeness claimed.